Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which satellite channel can show a jet stream in clear air?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Visible

B. Infrared only at night

C. Radar reflectivity

D. Water vapor

Correct: D

2. (2 points) Unsaturated air rises and cools at about:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. 9.8 °C per km

B. 2 °C per km

C. 6.5 °C per km always

D. 20 °C per km

Correct: A

3. (2 points) Which feature defines a supercell?

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. A warm front

B. A rotating updraft (mesocyclone)

C. A gust front

D. A single short-lived cell

Correct: B

4. (2 points) A hurricane with sustained winds of 115 mph is:

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Category 4

B. Category 1

C. Category 3

D. Category 2

Correct: C

5. (2 points) On Doppler velocity, green colors usually mean:

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Separate investigation observations:
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Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Heavy rain

B. Hail

C. Motion toward the radar

D. Motion away from the radar

Correct: C

6. (2 points) On a sounding, temperature and dew point lines close together indicate:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Very dry air

B. An inversion

C. Strong wind

D. Nearly saturated air

Correct: D

7. (2 points) In the dissipating stage of a single-cell storm:

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. The downdraft dominates

B. The updraft strengthens

C. The cloud is still growing

D. Rain has not begun

Correct: A

8. (2 points) Where are a hurricane's strongest winds?

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Far ahead of the storm

B. In the eyewall

C. In the eye

D. In the outer rain bands

Correct: B

9. (2 points) Why are cold, bright cloud tops on infrared images important?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. They indicate clear skies

B. They mark tall thunderstorms

C. They mark fog

D. They show surface temperature

Correct: B

10. (2 points) An environmental lapse rate between the moist and dry rates is:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Absolutely unstable

B. Always neutral

C. Conditionally unstable

D. Absolutely stable

Correct: C

11. (2 points) Interpret the adjacent opposite radial velocities cautiously.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: A tight adjacent inbound/outbound couplet is evidence consistent with rotation; geometry, resolution and quality control still matter. (2 points).

12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: Mixed or irregular scatterers consistent with lofted debris can support a tornado-debris interpretation when collocated with rotation; use corroborating evidence. (2 points).

13. (2 points) Calculate the surface temperature-dew-point spread.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: 5 C (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , C.

14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

answer: High reflectivity can indicate large or numerous precipitation targets, including hail; it does not directly establish a rotating ground-contact vortex. (2 points).

15. (2 points) A storm moves 25 km east in 30 minutes. Find mean speed.

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 25 km

answer: 50 km/h (2 points).

Numerical target 50; absolute tolerance 0.1. Accepted units: , km/h.

16. (2 points) A rain gauge records 35 mm in 0.5 hours. Calculate mean rain rate.

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 35 mm

answer: 70 mm/h (2 points).

Numerical target 70; absolute tolerance 0.14. Accepted units: , mm/h.

17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

answer: Colder cloud tops can be consistent with cloud growth to higher levels, but retrieval, cloud type and environmental temperature profile affect interpretation. (2 points).

18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

answer: A thunderstorm cold pool/outflow boundary is consistent with the observations; compare radar, temperature changes and timing rather than identifying it from one variable alone. (2 points).

19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

answer: Doppler radial velocity is only the component along the beam; perpendicular motion contributes little to that component. (2 points).

20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

answer: Antecedent moisture, infiltration, drainage, catchment area and routing affect runoff and flood response; evaluate rainfall intensity and basin conditions together. (2 points).

21. (2 points) Use Figure F1. Read the velocity difference between the adjacent west and east cells.

Same small storm area: collocated measurements

Region	Reflectivity	Radial velocity	Correlation
West cell	65 dBZ	-30 m/s	0.70
East cell	60 dBZ	+35 m/s	0.72
Background	25 dBZ	+5 m/s	0.98

F1. Synthetic dual-polarization radar panels (original synthetic teaching figure)

answer: 65 m/s (2 points).

Numerical target 65; absolute tolerance 0.195. Accepted units: , m/s.

22. (2 points) Use Figure F1. Explain why the juxtaposed velocity signs support investigating rotation rather than concluding that both cells are calm.

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

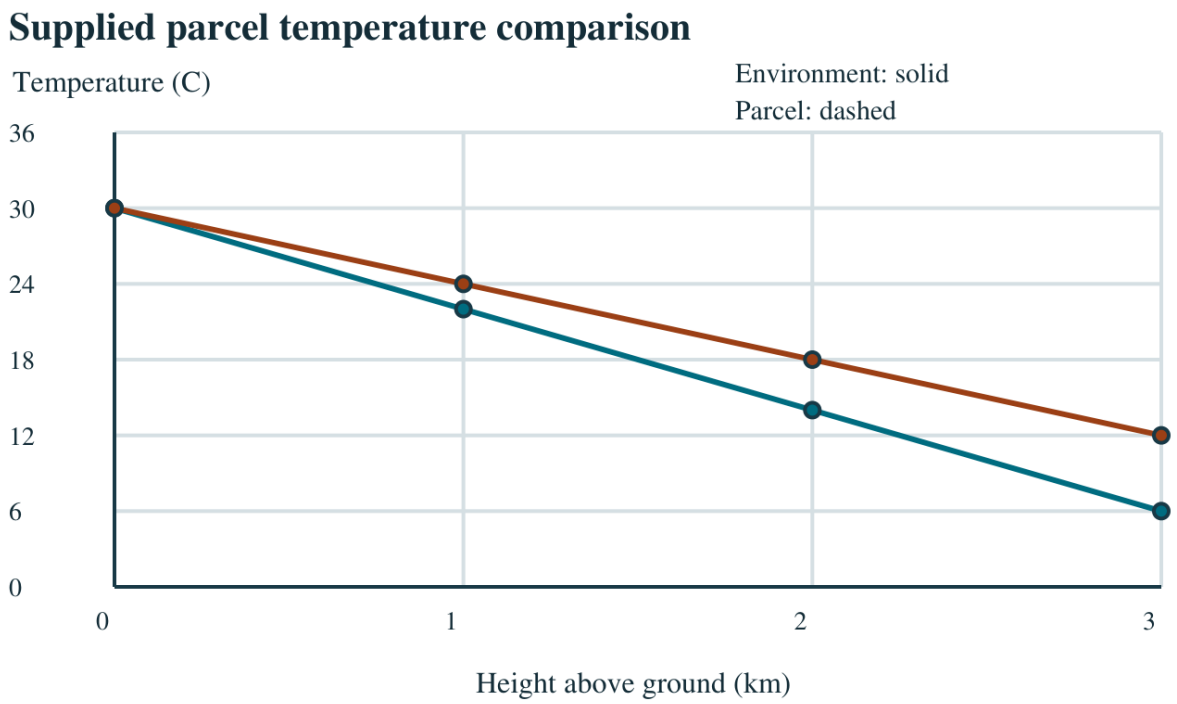
answer: The adjacent approaching and receding velocities form a couplet consistent with rotation; geometry, resolution and other evidence are needed for confirmation. (2 points).

23. (2 points) Use Figure F1. What extra interpretation does low correlation in the same area support, and what does it not prove by itself?

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

answer: It can support mixed/irregular scatterers such as lofted debris when collocated with rotation, but cannot independently prove a ground-contact tornado without corroborating evidence. (2 points).

24. (2 points) Use Figure F2. Read parcel-minus-environment temperature at 2 km.



F2. Parcel and environment profiles (original synthetic teaching figure)

answer: 4 C (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: , C.

25. (2 points) Use Figure F2. Under the stated simplification, is the parcel positively or negatively buoyant at 2 km? Explain.

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

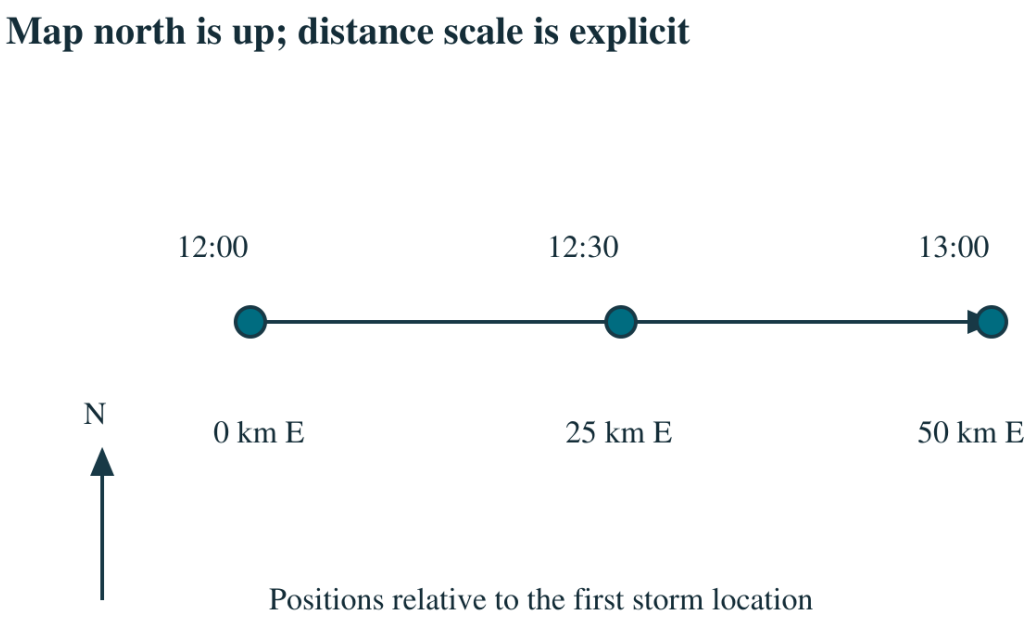
answer: Positively buoyant: the parcel is warmer and hence less dense than its environment at matched pressure under the stated simplification. (2 points).

26. (2 points) Use Figure F2. Why cannot this profile alone establish tornado occurrence?

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

answer: It does not supply all required storm and wind information, including shear/rotation, initiation and ground-level evidence; instability alone does not establish a tornado. (2 points).

27. (2 points) Use Figure F3. Calculate mean storm speed from 12:00 to 13:00.



F3. Storm-track map (original synthetic teaching figure)

answer: 50 km/h (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , km/h.

28. (2 points) Use Figure F3. Give the direction of the plotted motion.

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: East. (2 points).

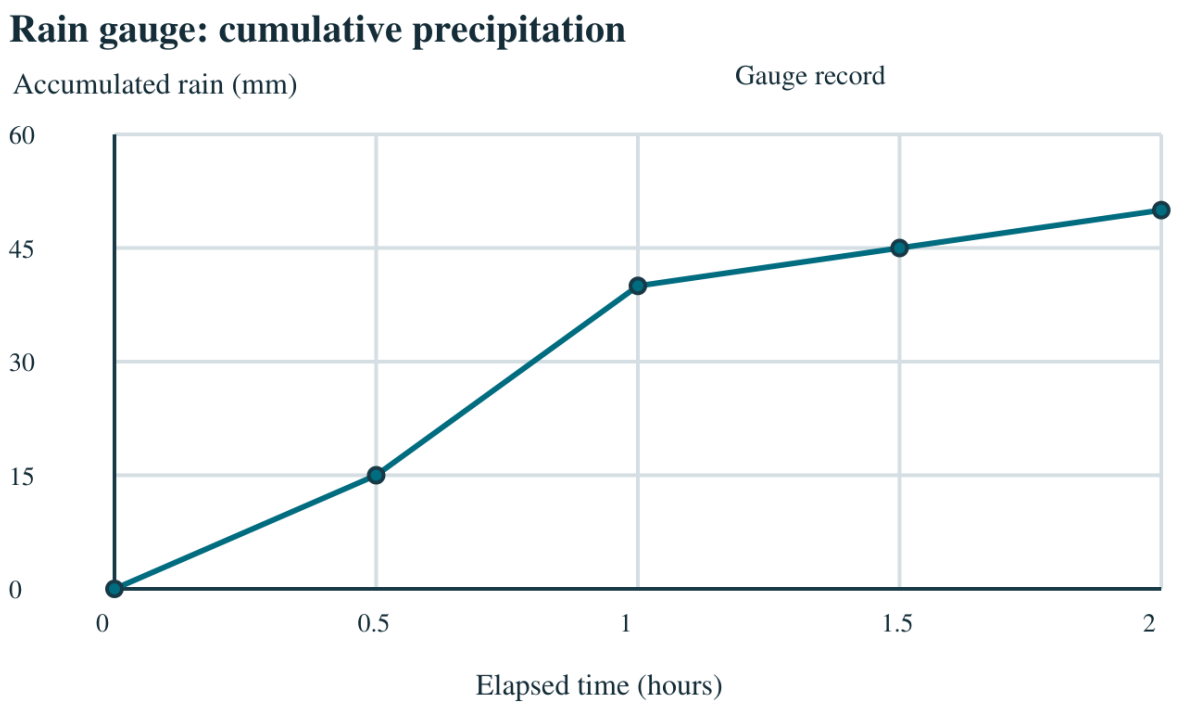
29. (2 points) Use Figure F3. If the motion continues unchanged, how many minutes after 13:00 would a point 25 km farther east be reached?

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: 30 minutes (2 points).

Numerical target 30; absolute tolerance 0.09. Accepted units: , minutes.

30. (2 points) Use Figure F4. Calculate the mean rain rate during the 0.5-to-1.0 h interval.



F4. Accumulated rainfall curve (original synthetic teaching figure)

answer: 50 mm/h (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , mm/h.

31. (2 points) Use Figure F4. Identify the interval with the steepest accumulation and explain what the slope measures.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.5 to1.0h; slope is rainfall rate, and this interval has the largest increase per unit time. (2 points).

32. (2 points) Use Figure F4. For an already saturated basin, explain why the steep interval deserves attention even if a later interval has a higher cumulative total.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: Short intense rainfall can create rapid runoff when infiltration capacity/storage is limited. Cumulative total and current intensity describe different aspects of flood response. (2 points).